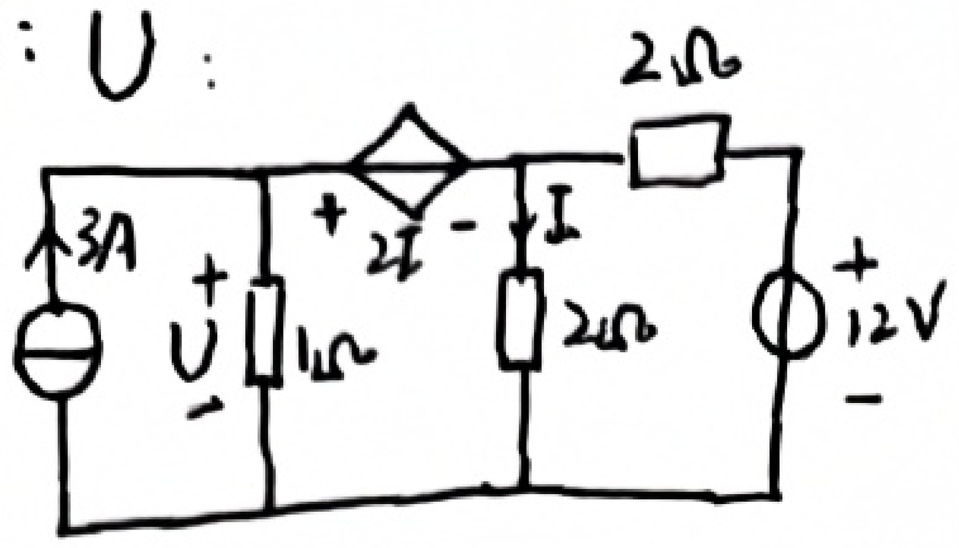
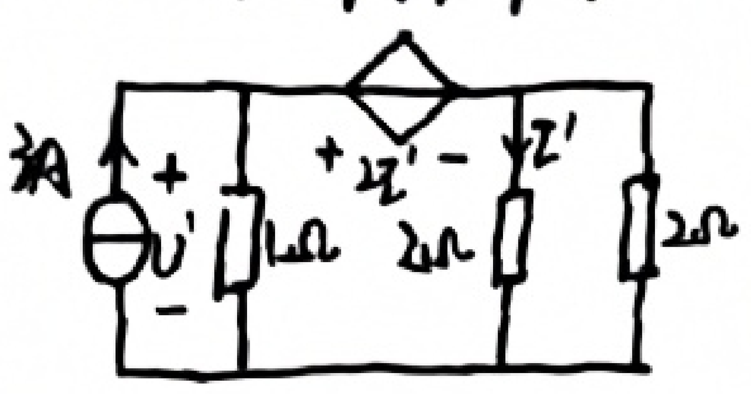
9月30日作业：　　作业②.

1. 叠加定理求：U：



① 3A电源单独作用.



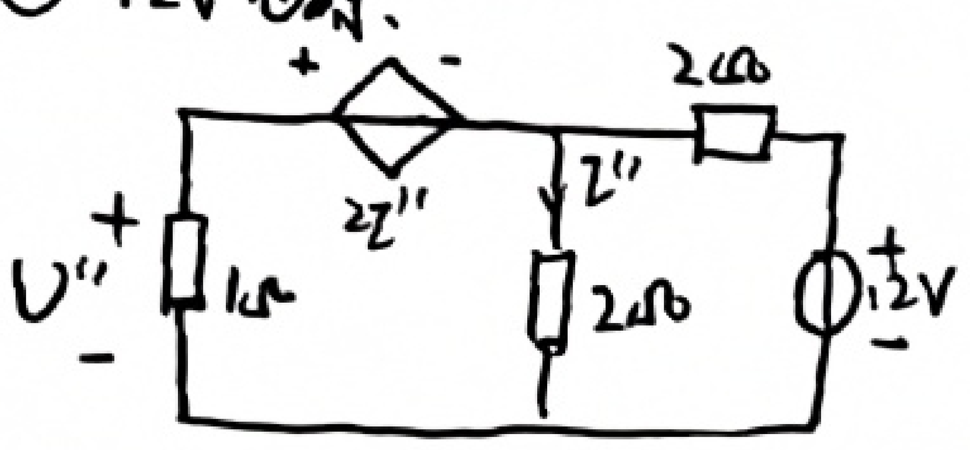
1Ω电阻上的 U'

$U'=2I'+2I'$

$\frac{U'}{1}+2I'=3$

即 $U'=2V$，$I'=0.5A$.

② 12V电源.



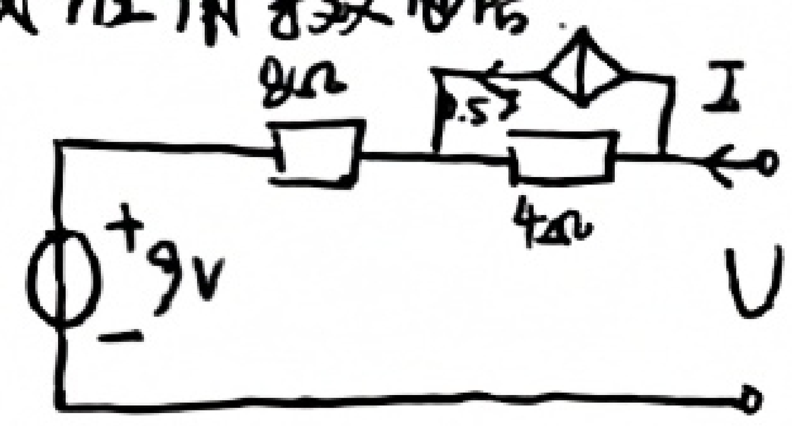
$U''=2I'+2I''$

$2I''=-2\times(I''+U'')+12$

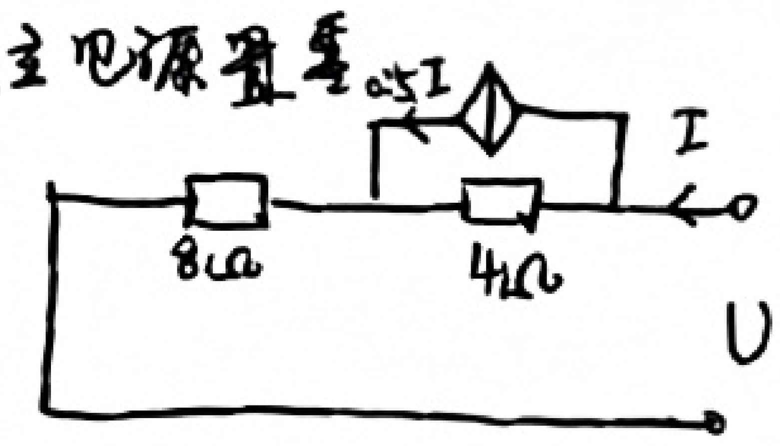
即 $U''=4V$　$I''=1A$

故 $U=U'+U''=6V$.

2. 戴维南等效电路：



独立电源置零



$U=4\times(I-0.5I)+8I$

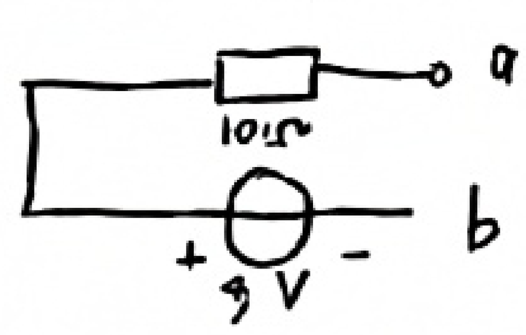
$R_0=\frac{U}{I}=10Ω$.

开路电压：当端开路时，I=0 即受控源 0.5I=0.

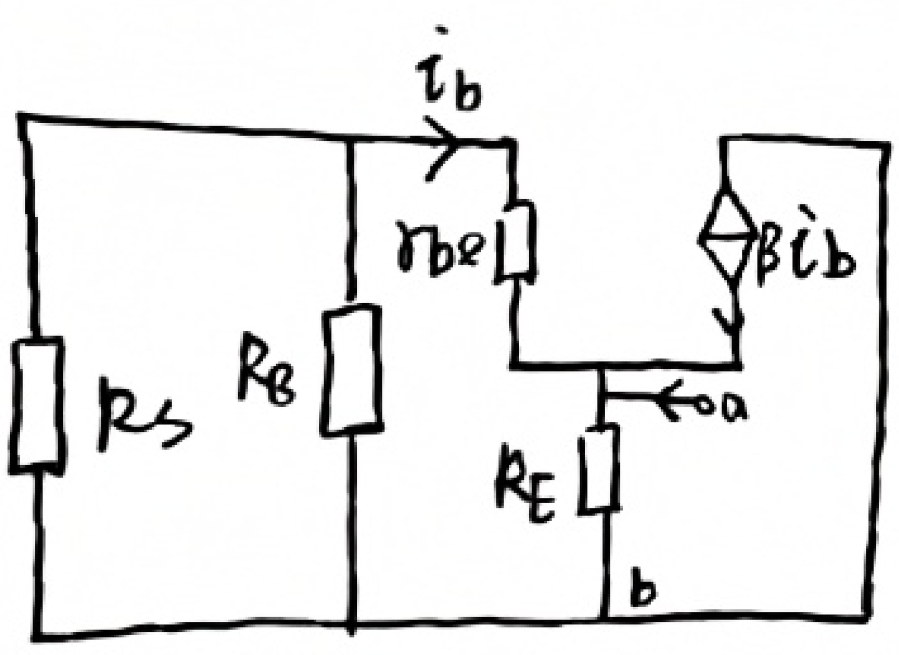
因此电路中只有9V电压源作用.

即 $U_{oc}=9V$

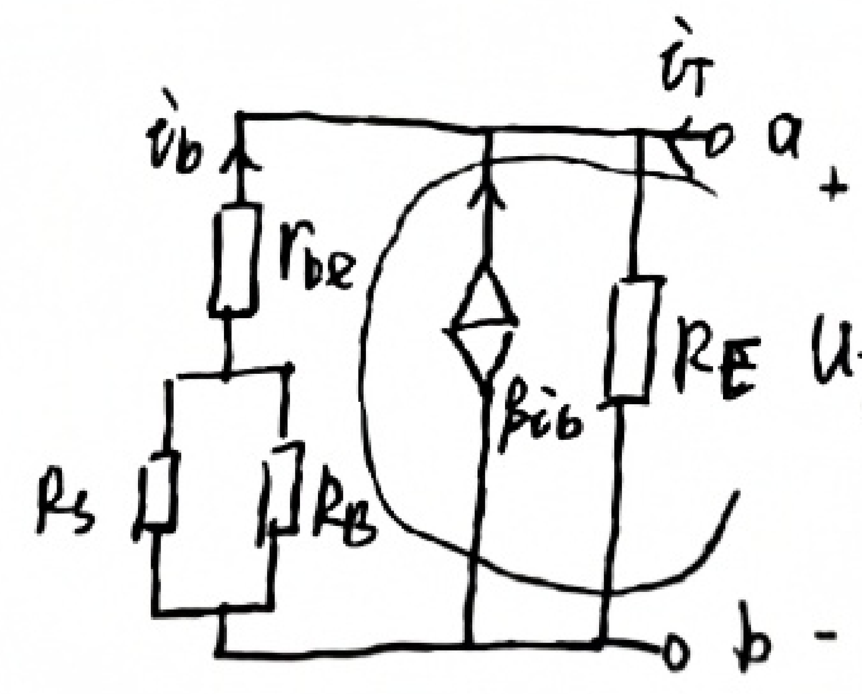
即等效电路为：



3. 从ab端看入的等效电阻.



⇒



假设 i_T 电流方向：

~~流过~~ 流过 R_E 的电流为：$i_E=i_T+i_b+\beta i_b$

$U_T=R_E\cdot i_E$

回路：$(R_S\|R_B+r_{be})i_b+U_T=0$

$\Rightarrow \cancel{U_T=}$

$$U_T=\frac{(R_S\|R_B+r_{be})R_E}{R_S\|R_B+r_{be}+(1+\beta)R_E}\cdot i_T$$

$$r_{ab}=\frac{U_T}{i_T}=\frac{(R_S\|R_B+r_{be})R_E}{R_S\|R_B+r_{be}+(1+\beta)R_E}$$